

How to Modify a Masterbuilt Digital Smoker to Work with Auber's Plug-n-Play PID Controller

*Auber Instruments
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Introduction

Electric smokers made by Masterbuilt Manufacturing LLC have two types of temperature-control modules:

- An analog controller with a temperature dial that is built on the heater plug (Figure 1, left).
- A digital control panel with digital display (Figure 1, right).

The analog control type can be simply plugged into the HEATER socket on Auber's PID controller. Set the analog dial to its maximum and let the Auber controller take over the temperature regulation.

However, smokers with digital control modules cannot be directly controlled by an Auber PID controller. Every time the smoker is powered up, the digital control module enters standby mode. To start heating, a user needs to press some buttons from the digital control panel. Some models won't start heating unless the sensor reads a temperature that is greater than 32°F. Therefore, the smoker's circuit must be modified to bypass the OEM digital control module so that the heating element can be controlled directly by an Auber's PID controller.

This document provides a simple solution on how to modify a digital smoker so that the smoker temperature can be controlled by Auber's plug-n-play PID controller.



Figure 1. A Masterbuilt electric smoker with analog controller (left), and a Masterbuilt electric smoker with digital control panel on the top (right).

Overview

Despite the differences among different models of digital smokers made by Masterbuilt, the wiring to the heating element is very similar. The electric circuit can be divided into four main sections.

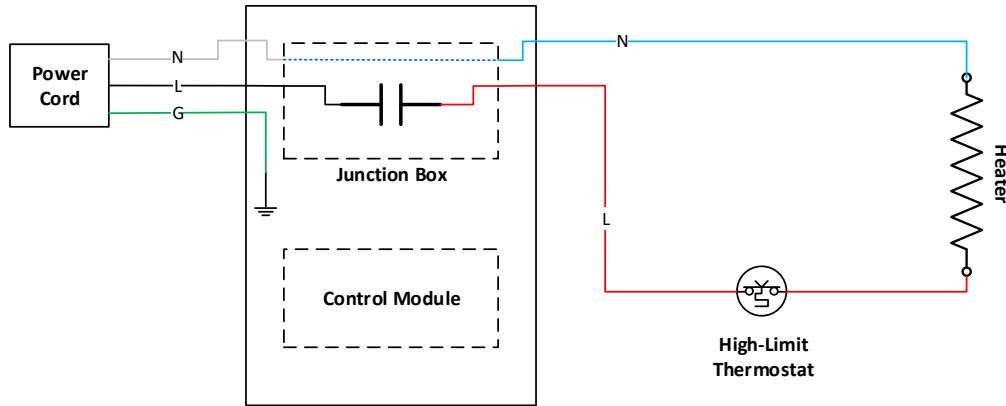
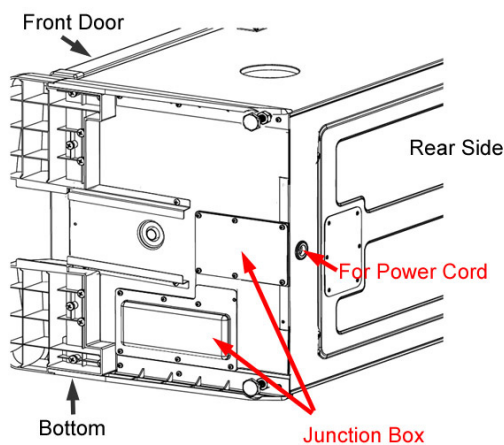
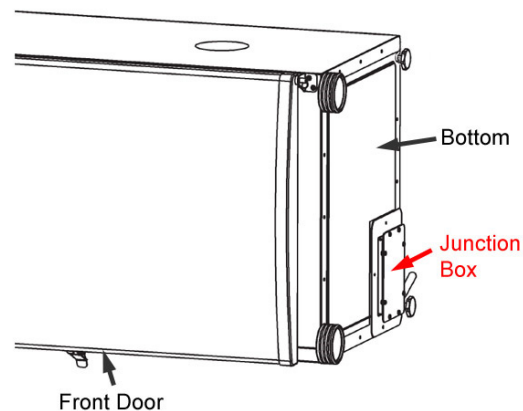


Figure 2. A general block diagram of a digital electric smoker.

1. **The junction box(es) that hold(s) the power cord and the main circuit board.** Typically located at the bottom of the smoker. Depending on the model, there may be two junction boxes on the bottom (Figure 2a) or just one junction box on the bottom (Figure 2b).



(a) Model # 20078715.



(b) Model # 20070106.

Figure 3. Two examples of the junction boxes. In some models (left, model # 20078715), there are two junction boxes on the bottom: one for the power cord and one for the circuit board. In some other models (right, model # 20070106), there is only one junction box on the bottom for the power cord and the circuit board.

2. **A digital control panel.** The control panel is typically located on the top of the smoker and is powered by a low voltage DC power.
3. **A high-limit thermostat.** This thermal cutoff thermostat, which is attached to the backwall on the smoker (Figure 4), will cut off power to the heating element if smoker temperature rises above 350°F.



Figure 4. High-limit thermostat.

4. **A heating element.** From the thermostat, the AC power goes to another junction box and connects to the heating element, which can be accessed from the back panel of the smoker.

Note: please be aware that not all models have junction boxes on the bottom. One of our customers who owns an old digital model reported that there is no junction box at the bottom of his smoker, instead the main control circuit board is located in the junction box on the back panel of the smoker.

Basically, there are two ways to modify the circuit in an electric smoker.

- Disconnect the heating element from the original circuit and attach a new power cord to the heating element.
- Disconnect the heating element from the original circuit and directly connect it to the original power cord. The OEM control module is bypassed and will not get power anymore.

The first method is relatively simple and straightforward.

In this document, we will only cover the second method, i.e., use the existing power cord and completely bypass the original control module. And we will use photos from a few different models to demonstrate how to modify the circuit in a digital smoker.

1. A 30 Inch Electric Smokehouse Smoker (model # 20070106).
2. A 30 Inch Digital Electric Smoker (model # 20072115).
3. A 30 Inch Digital Electric Smoker (model # 20078715).
4. A 40 Inch Digital Electric Smoker (model # 20072918).

Modification Procedures

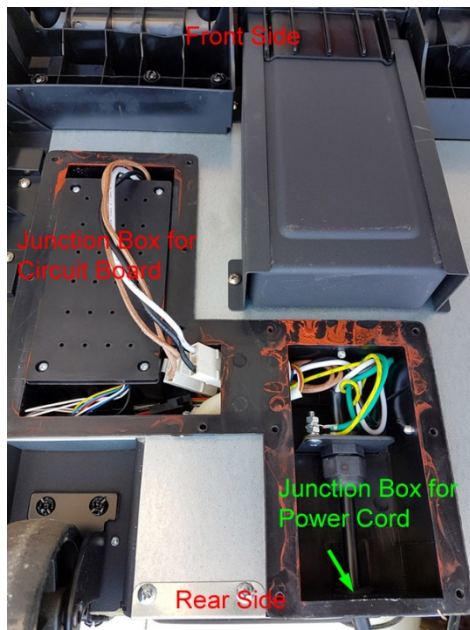
As we mentioned earlier, depending on the model there may be two junction boxes on the bottom (Figure 2(a)) or just one junction box on the bottom (Figure 2(b)).

- a. If there are two junction boxes on the bottom (Figure 2(a)), one for the 120 VAC power cord and one for the main circuit board, and the power cord is connected to the circuit board through a pair of connectors, **please refer to Section A below.**
- b. On some models, there is a single junction box on the bottom (Figure 2(b)). The 120 VAC main power goes to the circuit board directly. **If this is the case for your smoker, please refer to Section B below.**

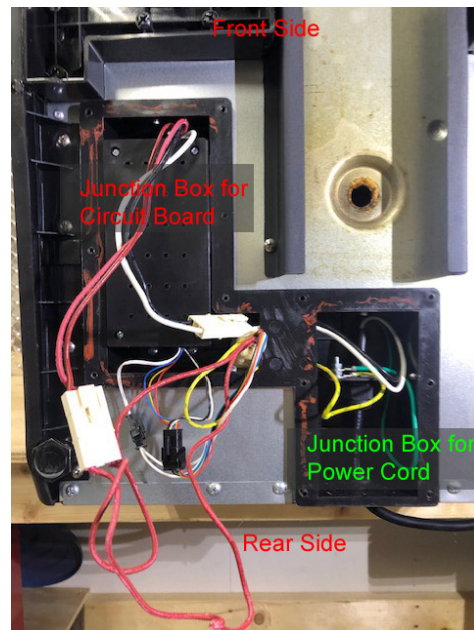
A. For Smokers with two junction boxes on the bottom

1) Remove the Cover Plates and Identify the Junction Boxes

Here are two examples of the junction boxes on the bottom of a smoker. The junction box on the lower right is where the power cord comes in. The junction box on the top left is where the circuit board is held.



(a) Model # 20072115



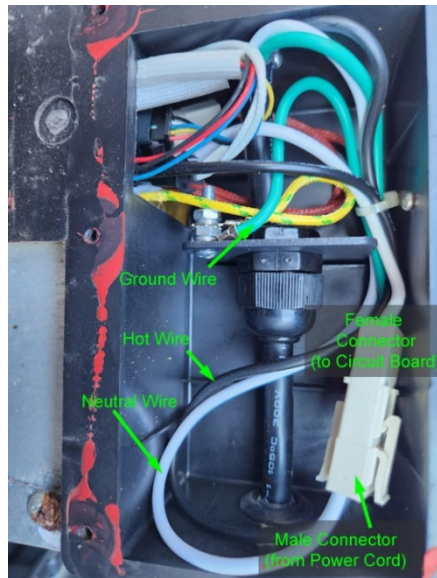
(b) Model # 20078715

Figure 5. Two examples of smokers with two junction boxes. In each image, the smoker is positioned with the power cord at the bottom and the front door facing upward.

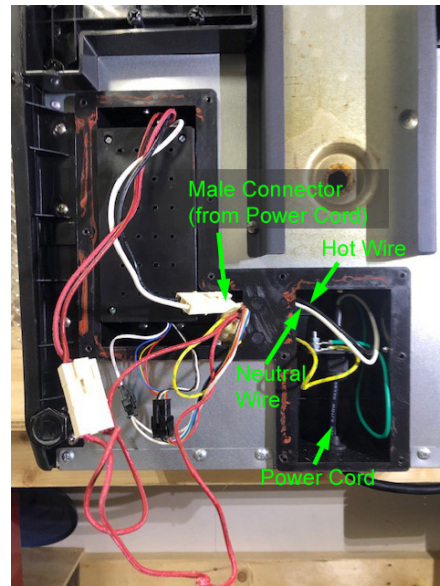
2) Identify the Wires

The power cord has three conductors: the black wire is the hot, the white wire is the neutral, and the green wire is the ground. The black wire and the white wire from the power cord are crimped into a three-pin male connector with the middle pin empty. It is plugged into a female connector which also has a pair of black wire and white wire that go to the circuit board. This supplies the 120 VAC power to the circuit board and to the heating element.

There is another pair of red wires with braided insulation comes out from the circuit board junction box. The color of the braiding may vary from model to model. These wires are terminated with the same type of three-pin male connector with the middle pin empty. The male connector is plugged to another female connector with the same braided wires which lead to the heating element. These two wires carry the AC power switched by a mechanical relay on the circuit board and supply the power to the heating element.

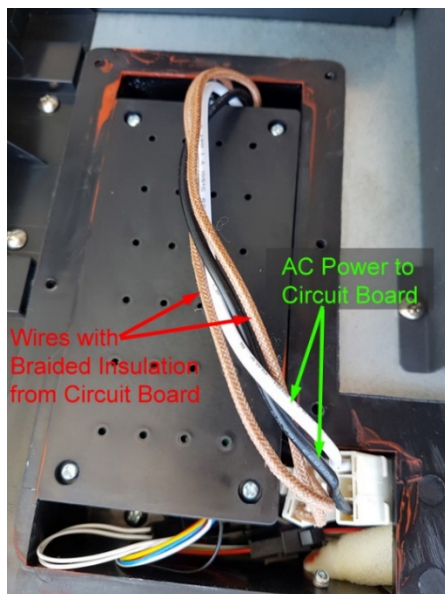


(a) Model # 20079015

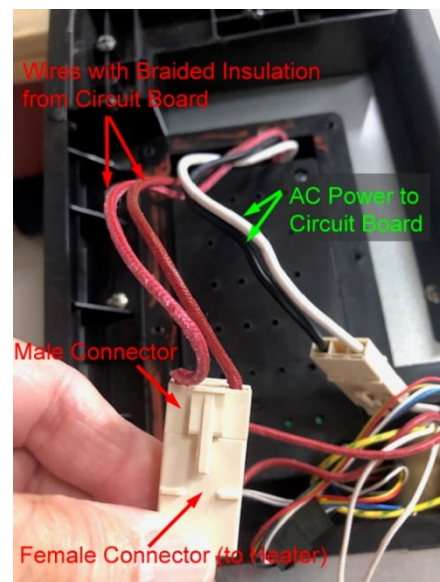


(b) Model # 20078715

Figure 6. Identify the hot wire, the neutral wire, and the male connector from the power cord.



(a) Model #20072115.



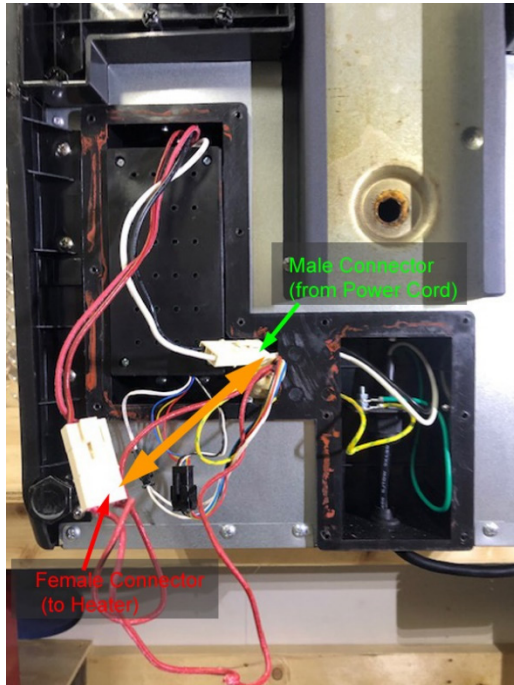
(b) Model #20078715.

Figure 7. The wires with braided insulation coming from the circuit board are terminated with a male connector. The corresponding female connector is attached to the wires leading to the heating element.

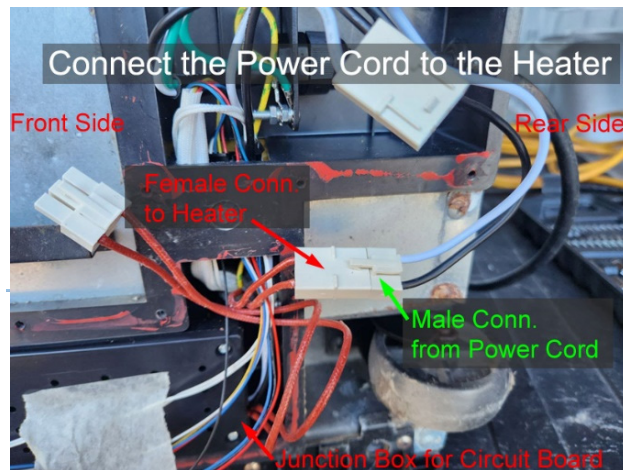
3) Modification

In these smokers, the modification is simple. You'll need to disconnect the two pairs of connectors. Identify the male connector (pointed by the green arrow) from the power cord and

the female connector (pointed by the red arrow) with braided wires that lead to the heating element as shown in Figure 8(a). Then, simply plug them together. The final connection should look like Figure 8(b).



(a)



(b)

Figure 8. (a) Identify the male connector attached to the power cord and the female connector attached to the heating element as pointed by the orange arrows. (b) Join these two connectors together.

B. For Smokers with a Single Junction Box on the Bottom

1) Open the Junction Box on the Bottom of the Smoker.

For smokers with one junction box on the bottom and another junction box on the back panel, open the one located on the bottom first.

Note: On some models, the cover of the junction box is held by sheet metal screws, which can be removed using a screw driver. But on some other models, the cover is held by rivets. You can use a 1/8" drill bit to drill out the rivets.



(a)



(b)

Figure 9. The junction box that holds the main control circuit board (indicated by the red circle) is located at the bottom of the smoker (model # 20070106).

2) Identify the Wires

The 120 VAC main power feeds into the circuit board and splits into two paths: (1) one path goes to a transformer, providing a low voltage DC power to the control panel; (2) the other path goes through a relay (controlled by the control panel) to power the smoker's heating element.

The actual wiring and the circuit board inside a junction box can vary from one model to another. We are showing pictures taken from three different models in Figure 10. However, we only need to identify three common components that are relevant to this modification. We have labelled these components with letters and arrows in different colors.

- A. AC Power-In.** The wires indicated by the **green arrows** are wires from power cord. These wires have smooth PVC plastic insulation. The black one is the hot wire and the white one is the neutral wire.
- B. Wires with Braided Insulation to the Heating element.** The wires indicated by the **red arrows** go to the heating element. These two wires have braided jacket, one of which connects directly to the heating element, the other one is attached to a relay. These wires are black in most models, but they could be in two different colors (e.g., the red wire and blue wire shown in Figure 6).

Note: Some models may have more than two wires with braided insulation. But the wires we are interested in should be thicker than the other braided cables.

- C. A Mechanical Relay.** The relay pointed by a **blue arrow** is controlled by the control panel in the smoker and is responsible for switching the power to the heating element. The

relay has two thick wires attached to it: (1) one is power-in wire with PVC insulation, and (2) the other one is a braided wire with the out-going power to the heating element.

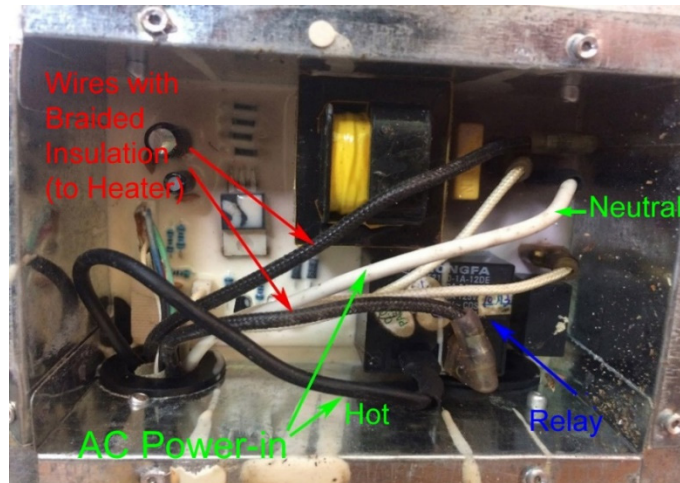


Figure 10 (a). The wiring of the main control board junction box of an older Masterbuilt model. The junction box is located on the back panel of the smoker.

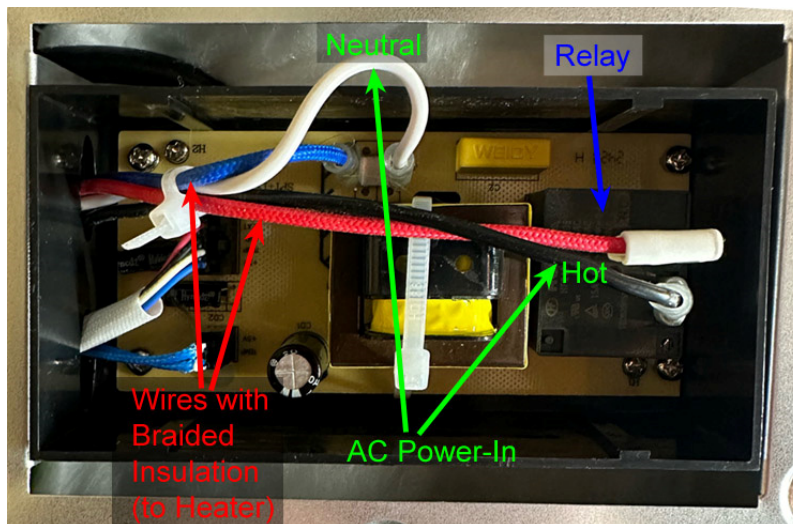


Figure 10 (b). The wiring of the junction box of a newer Masterbuilt (model # 20072918). The box is located at the bottom of the smoker.

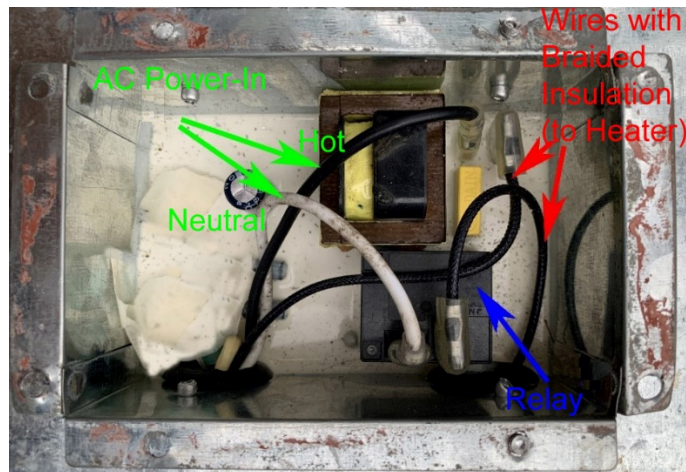


Figure 10 (c). The wiring in the junction box of a Masterbuilt 30 Inch Electric Smokehouse Smoker (model # 20070106). This junction box is located at the bottom of the smoker.

3) Modification

A simple way to modify the circuit is to join the two wires on the relay module with a piece of self-made jumper wire.

- A. Make a 2- to 3- inch-long jumper wire using 14 AWG stranded copper wire. Crimp 1/4" male spade connectors to both ends as shown in Figure 11.



Figure 11. Make a 14 AWG jumper wire terminated with 1/4-inch male spade connectors.

- B. Identify the two wires on the relay module. One is a PVC insulated wire from the power-in cord. The other is a wire with braided insulation that goes to the heating element. Unplug these two wires from the relay module. The arrows in Figure 12 indicate the wires that need to be unplugged.

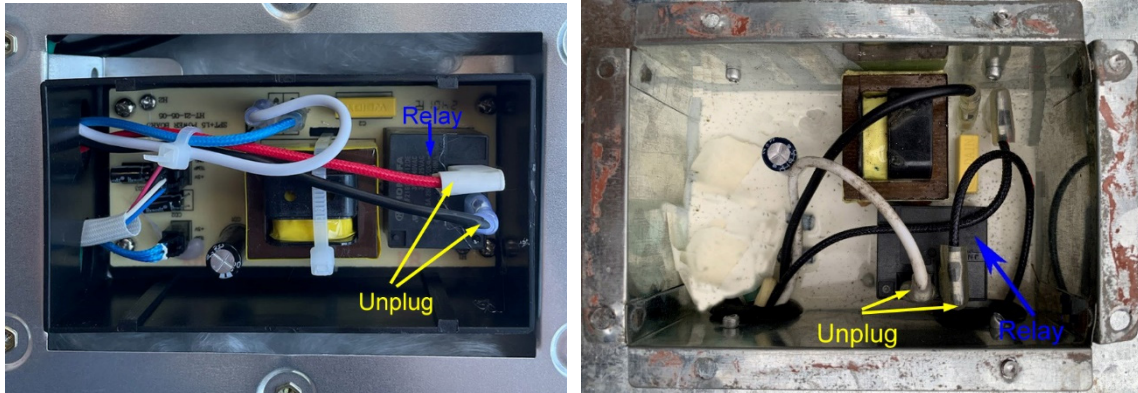


Figure 12. Identify the wires on the relay module to be unplugged.

- C. Join these two wires together with the jumper wire we just made as shown in Figure 13. Please refer to Figure 10 for its original wiring.

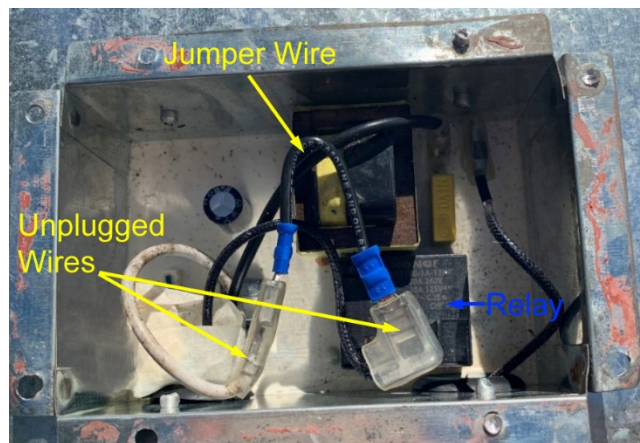


Figure 13. The two wires originally attached to the relay are joined together by the jumper wire (folded to fit the junction box).

The OEM control panel and the original relay module have now been bypassed. As a result, the control panel will no longer receive power, and the power to the heating element will no longer be switched by the relay module on the circuit board.

4) Cover the Junction Box

Cover the junction box with the original cover. If the cover was originally held in place by rivets, you need to rivet it back with 1/8" rivets and a rivet tool.

Use Auber's PID Controller

Simply plug the smoker's power cord into the "HEATER" socket on an Auber PID controller (such as WS-1211H, WSD-1200H, AW-1520H, or WST-1520H) and let the Auber controller regulate power to the heating element.

Note: Some Masterbuilt smokers may have more powerful heating elements (e.g., 1200W or 1500W). For these models, please choose our controller models capable of handling such power, such as AW-1520H, or WST-1520H. It is recommended to change the default proportional band setting (parameter "P") from 7 to 12. Other parameters ("I" and "D") can remain at their default values.



Figure 10. Use Auber WSD-1200GPH to control the temperature of a Masterbuilt smoker.

Alternative Solutions

The modification solution we provided in this document is simple and easy to reverse. Below is a list of other modification solutions that can be found online:

1) Some people want to have the option to use the original OEM controller when needed. In that case, you can add a switch between the two wires on the relay. Here is a link that shows how it is done.

<https://www.smokingmeatforums.com/threads/bypass-surgery-for-the-mes-30.273392/>

2) Some people still want to use the original controller as a thermometer to read the temperature. You can add an extra power cord to do it. Here is a link on how it is done.

<https://www.smokingmeatforums.com/threads/welcome-master-bradley-auber.288248/#post-1969335>

3) Additional reference.

<https://www.smokingmeatforums.com/threads/how-to-bypass-controller-on-mes-30-digital.136236/>

(END)

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